MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, ANALOG-TO-DIGITAL CONVERTERS WITH 3-STATE BUFFERED 10 BIT OUTPUTS, MONOLITHIC, SILICON

This amendment forms a part of MIL-M-38510/134, dated 6 May 1985, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.4, line 3: Delete "Case" and substitute "Ambient".

PAGE 6

TABLE I, Data blanking delay test, conditions column, line 2: Delete "$\overline{DR}$" and substitute "B & $\overline{C}$". In the minimum limits column: Delete "0.1" and substitute "0.2".

TABLE I, Data ready delay test, conditions column: Add " 6/". In the minimum limits column: Delete "0.1" and substitute "0.3".

TABLE I, notes: Add " 6/ The data ready output should not be used to latch data. The MSB may tristate before the low to high transition of the $\overline{DR}$ output."

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

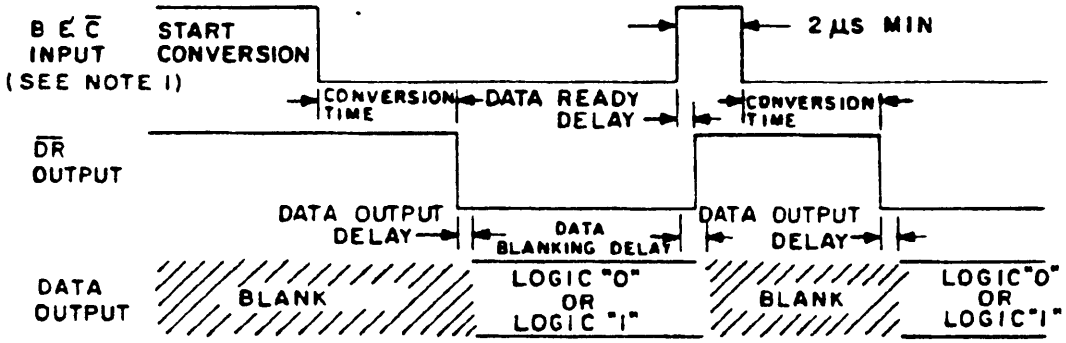
Replacement page	Page replaced
11	Reprinted without change
12	12

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:32Z
Check the source to verify that this is the current version before use.

PAGE 10

Figure 4: Delete and substitute new figure 4 as printed below.



TIMING & CONTROL SEQUENCE

NOTES: 1. The B and $\overline{C}$ start conversion pulse width shall be 2 µs minimum.

2. Data ready delay is 1.5 µs maximum.

FIGURE 4. Timing diagram.

PAGE 31

TABLE III, tests 835 through 854, minimum limits column: Delete "0.1" and substitute "0.2". For test 856, minimum limits column: Delete "0.1" and substitute "0.3".

PAGE 33

4.5.2, delete and substitute new paragraph as follows:

"4.5.2 Life test and burn-in cooldown procedure. When devices are measured at 25°C following application of steady-state life or burn-in test condition, they shall be cooled to within 10°C of their power stable condition at room temperature prior to the removal of the bias."

6.2.g, line 1: Delete " special carriers, ". Also delete the last sentence in its entirety.

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - AR, SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1003)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:32Z
Check the source to verify that this is the current version before use.

Analog Input		Digital Output
Unipolar	Bipolar	MSB LSB 1/
0.0098	-4.9951	00 0000 0000 2/
4.9902	-0.0146	01 1111 1111
5.0000	-0.0049	10 0000 0000
5.0098	+0.0049	
9.9902	+4.9854	11 1111 1111

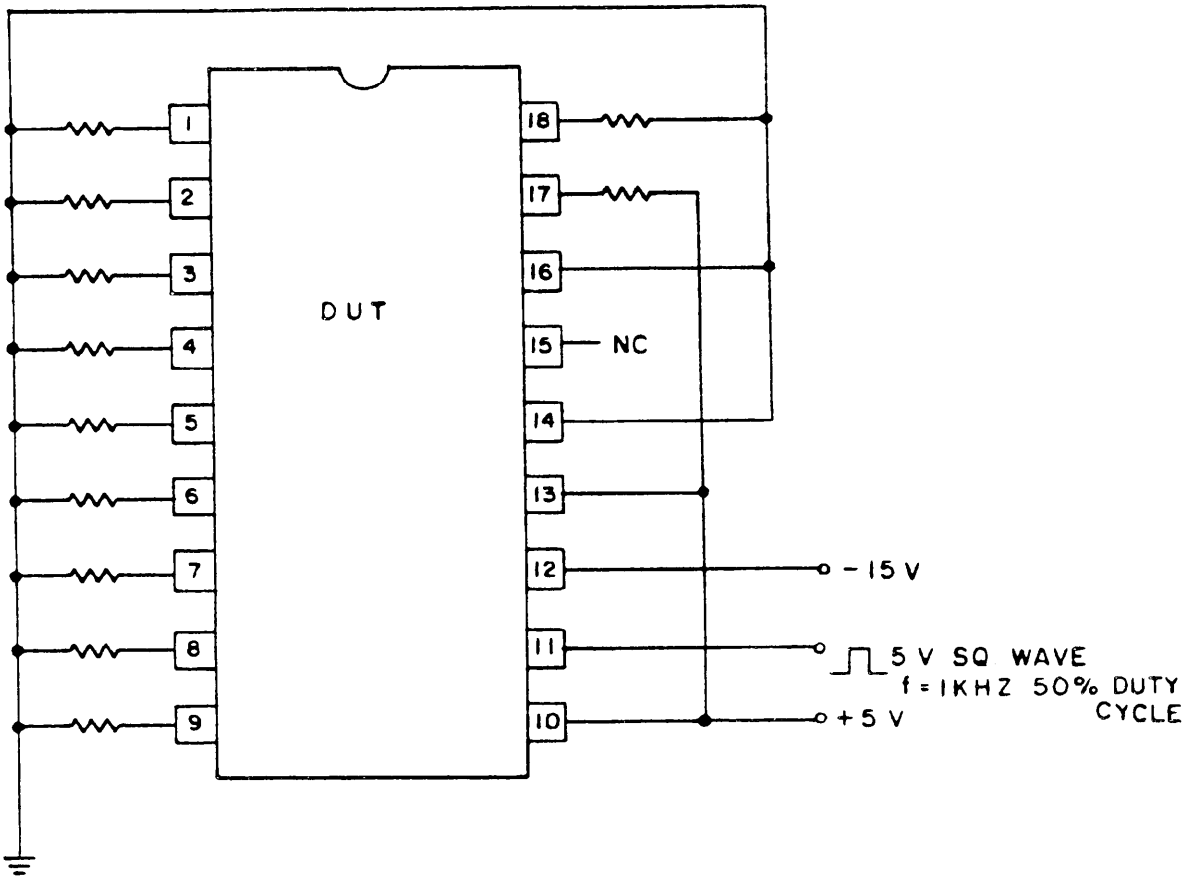
1/ 1 LSB - 9.766 mV

2/ Ø represents the transition point between two adjacent code-words (ie: 00 0000 0000 represents the transition between code-words 00 0000 0000 and 00 0000 0001, and ØØ ØØØØ ØØØØ represents the transition between code words 01 1111 1111 and 10 0000 0000).

FIGURE 5. Digital output codes.

Reprinted without change.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:32Z
Check the source to verify that this is the current version before use.



NOTES:
1. All resistors 3.0 K 1% 1/4W MF.
2. I_{CC} (+5 V) ∿ 1 Amp/100 devices.
   I_{EE} (-15 V) ∿ 2 Amp/100 devices.
3. 10 µf caps - one per supply.

FIGURE 6. Test circuit, burn-in and steady state life test.

Supersedes page 12 of MIL-M-38510/134 of 6 May 1985.

Source: https://assist.dla.mil -- Downloaded: 202610-21T13:32Z
Check the source to verify that this is the current version before use.